

```

#include<stdio.h>

int mutex=1, full=0, empty=3, x=0;

int wait(int s)
{
    return --s;
}

int signal(int s)
{
    return ++s;
}

int main()
{
    int ch;

    while(1)
    {
        printf("\n1.Producer\n2.Consumer\n3.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1:
                if((mutex==1)&&(empty!=0))
                {
                    mutex=wait(mutex);
                    full=signal(full);
                    empty=wait(empty);
                    x++;
                    printf("Produced Item %d\n",x);
                    mutex=signal(mutex);
                }
                else
                    printf("Buffer Full\n");
                break;

            case 2:
                if((mutex==1)&&(full!=0))
                {
                    mutex=wait(mutex);
                    full=wait(full);
                    empty=signal(empty);
                    printf("Consumed Item %d\n",x);
                    x--;
                }
                else
                    printf("Buffer Empty\n");
                break;

            case 3:
                return 0;
        }
    }
}

```

```

        mutex=signal(mutex);
    }
    else
        printf("Buffer Empty\n");
    break;

    case 3:
        return 0;
    }
}
}
}

```

OUTPUT:

1.Producer

2.Consumer

3.Exit

1

Produced Item 1

1.Producer

2.Consumer

3.Exit

1

Produced Item 2

1.Producer

2.Consumer

3.Exit

1

Produced Item 3

1.Producer

2.Consumer

3.Exit

2

Consumed Item 3

1.Producer

2.Consumer

3.Exit

2

Consumed Item 2

1.Producer

2.Consumer

3.Exit

3